

Unbounded Rotation Blocks Destroy the Operator-Norm Absorption Limit

Counterexample packet for arXiv:1511.02329

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Status: candidate counterexample, likely valid. The unconditional operator-norm extension of Theorem 1.2 to unbounded generators is false, even for a skew-adjoint generator and an orthogonal projection on a Hilbert space. The example still converges strongly to the natural compressed semigroup, isolating a purely high-frequency failure of operator-norm convergence.

1 Source and exact open problem

The source is J. Glück, *A note on approximation of operator semigroups*, arXiv:1511.02329, later *Archiv der Mathematik* 106 (2016), 265–273 [1]. For bounded A and a bounded projection P , with $Q = I - P$, its Theorem 1.2 proves

$$e^{t(A+zP)} \longrightarrow e^{tQAQ}Q \quad (\operatorname{Re} z \rightarrow -\infty)$$

in operator norm for every $t > 0$, uniformly on compact subintervals of $(0, \infty)$. At the bottom of source page 2 and top of page 3, the paper states that it is unclear what happens for unbounded generators and explicitly leaves the unbounded-generator case as an open problem.

As the generator A is bounded it is natural to ask whether the limit exists even in the operator norm topology. In Section 3 at the end of this note we will prove that this is indeed true. Our focus, however, is on another question, namely whether the limit $\lim_{k \rightarrow \infty} e^{t(A-kP)}$ always exists if A is bounded. Our main result gives an affirmative answer to this question:

Theorem 1.2. *Let A be a bounded linear operator and P a bounded linear projection on a complex Banach space X . Define $Q := 1 - P$ and let $z \in \mathbb{C}$. For all $t > 0$ we have*

$$e^{t(A+zP)} \rightarrow e^{tQAQ} \quad \text{as } \operatorname{Re} z \rightarrow -\infty$$

with respect to the operator norm, and the convergence is uniform with respect to t on compact subsets of $(0, \infty)$.

The proof of Theorem 1.2 which we present in Section 2 uses only elementary methods, but it requires a careful analysis of the spectral properties of $A + zP$.

In fact we will obtain a bit more information about the convergence in the above theorem, including an explicit estimate (for undefined notation we refer to the end of the introduction):

Remark 1.3. For t in any fixed compact subset of $(0, \infty)$, the convergence in Theorem 1.2 has at least a linear rate.

More precisely, the following holds: Let $0 < T_1 < T_2$ and define $R := 2(\|A\| + \delta)\|P - Q\|$, where $\delta > 0$ is any number sufficiently large to ensure that R is strictly larger than the spectral radius of QAQ . Then for every $z \in \mathbb{C}$ with $\operatorname{Re} z < -2R$ and all $t \in [T_1, T_2]$ we have

$$\|e^{t(A+zP)} - e^{tQAQ}\| \leq C_1 e^{T_1 \operatorname{Re} z} + C_2 \frac{1}{|z| - R},$$

where $C_2 := Re^{T_2 R} \frac{\|A\| + \delta}{\delta} \|P\| \sup_{|\lambda|=R} \|I + AQ \mathcal{R}(\lambda, QAQ)\|$ and $C_1 := \frac{Re^{T_1 R}}{\delta}$; here, $\mathcal{R}(\lambda, QAQ)$ denotes the resolvent of QAQ at λ . Note that C_2 can be further estimated by means of the Neumann series if we chose $\delta > 0$ sufficiently large such that even $R > \|QAQ\|$.

A few further remarks are in order.

Remarks 1.4. (a) Using complexifications one immediately obtains an analogues result for real Banach spaces.

(b) The operator family $(e^{tQAQ})_{t \geq 0}$ is a (norm continuous) C_0 -semigroup on $QX = \ker P$. Hence, the semigroup $(e^{t(A+zP)})_{t \geq 0}$ converges to 0 on the range of P and to a semigroup with generator QAQ on the kernel of P .

(c) Using the power series expansion of the exponential function one obtains several ways to write down the limit semigroup; in fact, one has

$$e^{tQAQ}Q = e^{tQA}Q = Qe^{tAQ} = Qe^{tQAQ}Q$$

for every $t \in (0, \infty)$.

(d) The assertion of Theorem 1.2 is obviously false for $t \leq 0$; just consider $X = \mathbb{C}$, $A = 0$ and $P = 1$ to see this.

(e) A glance at the proof of Theorem 1.2 in the next section reveals that one can show similar results for analytic functions f other than $\exp$, provided that f satisfies certain decay properties. Since our focus is on operator semigroups, we shall not discuss this in detail.

It seems to be unclear what happens to the assertion of Theorem 1.2 if we consider C_0 -semigroups with *unbounded* generator A . Of course one expects that some additional conditions are necessary in this case, since otherwise it might happen

Figure 1: Source PDF page 2: Theorem 1.2, its operator-norm assertion, and the beginning of the unbounded-generator question.

that QAQ has only a very small domain which need not be dense in the range of Q ; moreover, QAQ might not even be closed. Concerning the related Theorem [1.1](#) we also point out that the strong limit $\lim_{k \rightarrow \infty} (e^{\frac{t}{k}A}Q)^k$ never exists for all bounded linear projections Q on X unless A is bounded [\[6\]](#) Theorem 2.1]. In any case, the proof of Theorem [1.2](#) which we present below relies heavily on the boundedness of A . We therefore leave it as an open problem to analyse the case of unbounded generators.

Before we give a proof of our main result in the next section, let us briefly fix some notation. If X is a complex Banach space, then we denote by $\mathcal{L}(X)$ the space of all bounded linear operators on X . The spectrum of an operator $A \in \mathcal{L}(X)$ is denoted by $\sigma(A)$ and for every $\lambda \in \mathbb{C} \setminus \sigma(A)$, $\mathcal{R}(\lambda, A) := (\lambda - A)^{-1}$ denotes the resolvent of A at λ .

Figure 2: Source PDF page 3: continuation and explicit open-problem sentence.

2 Proof intuition

The bounded theorem controls all spectral modes using a constant depending on $\|A\|$. An unbounded generator has no such global frequency ceiling. We exploit this with a direct sum of planar rotations of frequencies $1, 2, 3, \dots$ and damp the first coordinate at rate k .

For each fixed rotation block, letting $k \rightarrow \infty$ suppresses the first coordinate and leaves the second coordinate fixed. Uniform contractivity then gives strong convergence on the Hilbert direct sum. Operator norm can choose a different block for each k : at frequencies $n \gg k$, fast rotation averages the damping across both coordinates, so even the nominally undamped second coordinate has size about $e^{-kt/2}$. The proposed limit Q leaves that coordinate unchanged. Their distance is therefore almost one.

3 Construction and theorem

Let

$$\mathcal{H} = \ell^2(\mathbb{N}; \mathbb{C}^2), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and define

$$\mathcal{D}(A) = \left\{ x = (x_n) : \sum_{n \geq 1} n^2 \|x_n\|^2 < \infty \right\}, \quad (Ax)_n = nJx_n.$$

Thus A is an unbounded skew-adjoint operator and $(e^{tA}x)_n = e^{tnJ}x_n$ is a unitary C_0 -group. Define the orthogonal projections, blockwise, by

$$P(a, b) = (a, 0), \quad Q(a, b) = (0, b).$$

Theorem 1. *For every $k > 0$, $A - kP$ generates a contraction semigroup on $\mathcal{H}$. For every $t > 0$ and $x \in \mathcal{H}$,*

$$e^{t(A-kP)}x \longrightarrow Qx \quad (k \rightarrow \infty).$$

However, for every fixed $t > 0$,

$$\left\| e^{t(A-kP)} - Q \right\| \geq 1 - e^{-kt/2}, \quad \liminf_{k \rightarrow \infty} \left\| e^{t(A-kP)} - Q \right\| \geq 1. \quad (1)$$

Consequently $e^{t(A-kP)}$ has no operator-norm limit as $k \rightarrow \infty$. Moreover, Q is zero on a dense domain in $Q\mathcal{H}$ and its closure is the zero generator, so $Q = e^{tQ}Q$ is precisely the natural compressed semigroup predicted by Theorem 1.2.

Proof. Both P and Q preserve $\mathcal{D}(A)$. If $x \in \mathcal{D}(A) \cap Q\mathcal{H}$, then each $x_n = (0, b_n)$ and $QnJx_n = 0$. Hence $QnJx_n = 0$ on the dense subspace $\mathcal{D}(A) \cap Q\mathcal{H}$ of $Q\mathcal{H}$, and its closure is zero on all of $Q\mathcal{H}$.

The n th block of $A - kP$ is

$$M_{n,k} = \begin{pmatrix} -k & -n \\ n & 0 \end{pmatrix}.$$

If $v'(t) = M_{n,k}v(t)$, then

$$\frac{d}{dt} \|v(t)\|^2 = 2 \operatorname{Re} \langle M_{n,k}v(t), v(t) \rangle = -2k|v_1(t)|^2 \leq 0.$$

Thus every block exponential is a contraction. Their Hilbert direct sum is a strongly continuous contraction semigroup, and its generator is $A - kP$.

We first prove strong convergence. Fix n . Once $k > 2n$, put

$$\delta = \sqrt{k^2/4 - n^2}, \quad \lambda_{\pm} = -k/2 \pm \delta.$$

The exact two-by-two exponential is

$$e^{tM_{n,k}} = \frac{e^{t\lambda_+} + e^{t\lambda_-}}{2} I + \frac{e^{t\lambda_+} - e^{t\lambda_-}}{2\delta} \left(M_{n,k} + \frac{k}{2} I \right). \quad (2)$$

As $k \rightarrow \infty$, $\lambda_+ \rightarrow 0$, $\lambda_- \rightarrow -\infty$, $k/(2\delta) \rightarrow 1$, and $n/\delta \rightarrow 0$. Formula (2) therefore gives

$$e^{tM_{n,k}} \longrightarrow Q_2 := \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

for every fixed n and $t > 0$. Hence convergence holds on finitely supported vectors. Since both $e^{t(A-kP)}$ and Q are contractions, a finite-support approximation gives convergence for every $x \in \mathcal{H}$.

It remains to show that this convergence is not uniform on the unit ball. Fix $k, t > 0$, and let $n > k/2$. With $\omega = \sqrt{n^2 - k^2/4}$, the underdamped formula gives

$$e^{tM_{n,k}} e_2 = e^{-kt/2} \begin{pmatrix} -\frac{n}{\omega} \sin(\omega t) \\ \cos(\omega t) + \frac{k}{2\omega} \sin(\omega t) \end{pmatrix}. \quad (3)$$

As $n \rightarrow \infty$, $n/\omega \rightarrow 1$ and $k/(2\omega) \rightarrow 0$, so uniformly in the oscillating phase,

$$\|e^{tM_{n,k}} e_2\| \longrightarrow e^{-kt/2}.$$

Let $f_n \in \mathcal{H}$ be the unit vector equal to e_2 in block n and zero elsewhere. Since $Qf_n = f_n$, the reverse triangle inequality yields

$$\left\| e^{t(A-kP)} - Q \right\| \geq \|e^{tM_{n,k}} e_2 - e_2\| \geq 1 - \|e^{tM_{n,k}} e_2\|.$$

Letting $n \rightarrow \infty$ proves the first inequality in (1), and then $k \rightarrow \infty$ proves the second. Any operator-norm limit would have to equal the already established strong limit Q , so no operator-norm limit exists. $\square$

4 Verification, novelty, and scope

Verification. The proof uses exact matrix exponentials. The included regression script checks fixed-mode convergence, high-frequency asymptotics, the lower bound, and contractivity over several parameter ranges. It is not part of the proof.

Bounded novelty search. The four run indexes were searched for arXiv:1511.02329, absorption semigroups, unbounded generators, projection damping, rotation blocks, and operator-norm convergence. Exact-title, exact-phrase, author/id, formula, quantum-Zeno, and close-variant arXiv/web searches through 27 August 2026 found generic references to the source but no exact answer. OpenAlex and Semantic Scholar citation metadata for the source DOI returned no indexed citing works. Novelty is plausible, not certified.

Scope and review. This completely disproves the unconditional operator-norm extension of Theorem 1.2 to arbitrary unbounded generators. It does not classify useful additional assumptions (analyticity, compact resolvent control, invariant domains, or spectral cutoffs) that restore convergence, so the source’s broad request to analyse the unbounded case remains open beyond this negative boundary result. Human review should check formulas (2)–(3) and the finite-support density argument.

References

- [1] J. Glück, *A note on approximation of operator semigroups*, Arch. Math. (Basel) **106** (2016), 265–273; [arXiv:1511.02329](#).